

Test

0.1 Krylov subspace

A Krylov subspace $\mathcal{K}_m(A, v)$ of $\mathbb{R}^n$ is spanned by the vectors

$$\mathcal{K}_m(A, v) \equiv \text{span}(v, Av, A^2v, \dots, A^{m-1}v)$$

Algorithm 1 Arnoldi Iterations [Saad2003]

Require: Choose v_1 s.t. $\|v_1\| = 1$

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for  $j = 1, 2, \dots, m$  do
  for  $i = 1, 2, \dots, j$  do
     $h_{i,j} \leftarrow (Av_j, v_i)$ 
  end for
   $w_j \leftarrow Av_j - \sum_{i=1}^j h_{i,j} v_i$ 
   $h_{j+1,j} \leftarrow \|w_j\|_2$ 
  if  $h_{j+1,j} = 0$  then
    Stop
  end if
   $v_{j+1} \leftarrow \frac{w_j}{h_{j+1,j}}$ 
end for
```
